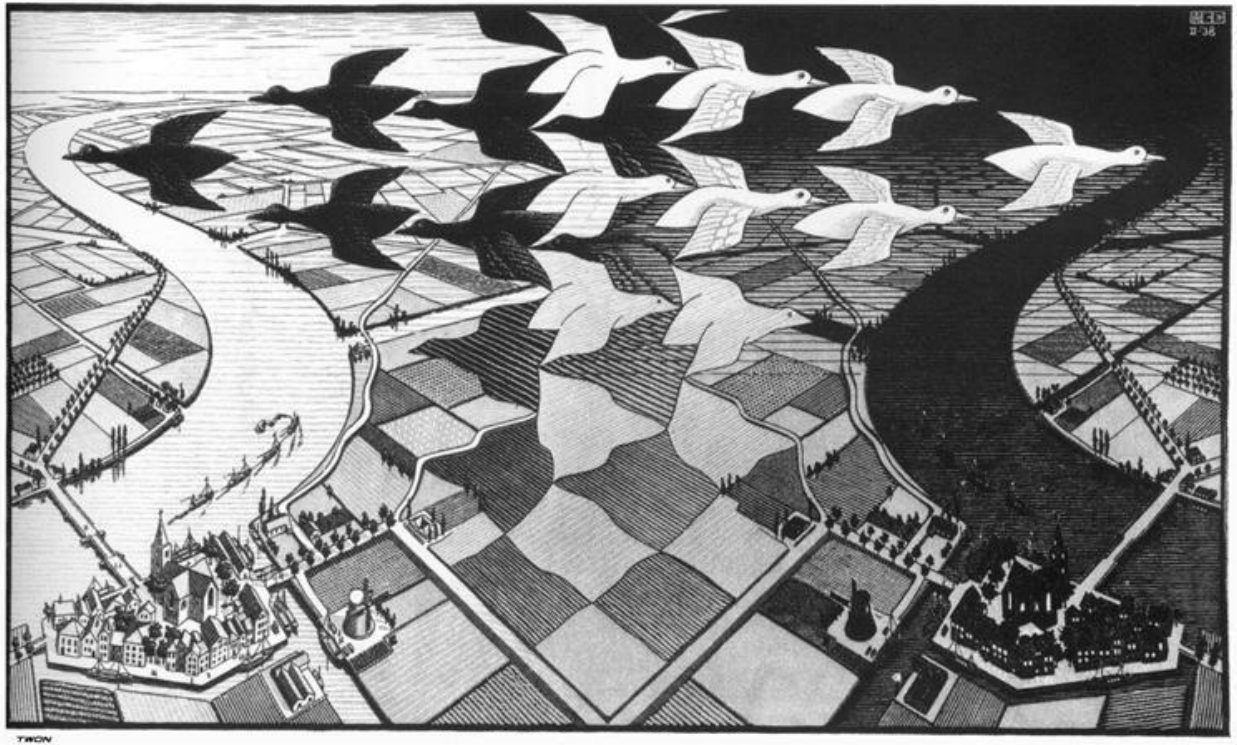
Studying predictive coding in the brain



M. C. Escher,
Day and Night (1938)

Exercise 1.1 | 16 September 2026

Eduarda Streit Morsch and Elizabeth Amadei

Published: 12 September 2016

Experience-dependent spatial expectations in mouse visual cortex

Aris Fiser, David Mahringer, Hassana K Oyibo, Anders V Petersen, Marcus Leinweber & Georg B Keller

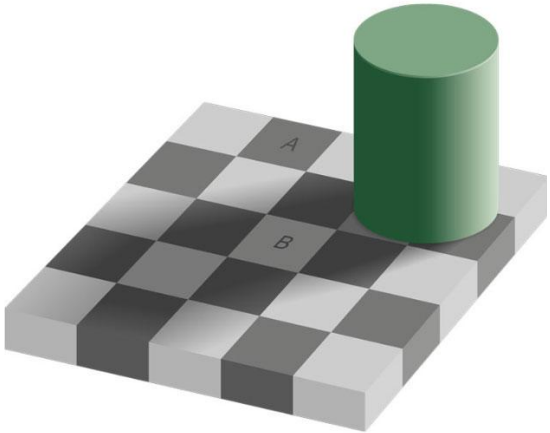
Nature Neuroscience **19**, 1658–1664 (2016) | Cite this article

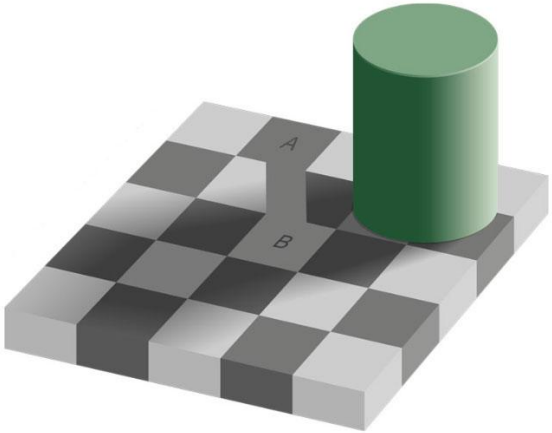
16k Accesses | **107** Citations | **37** Altmetric | Metrics

The case of the dress



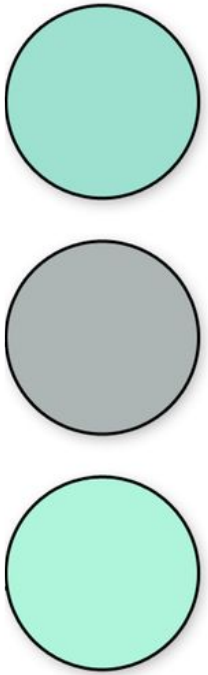
Team gold&white? Team blue&black?





Team A and B different colors? A and B the same colors?

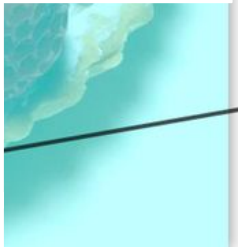
The case of the dress

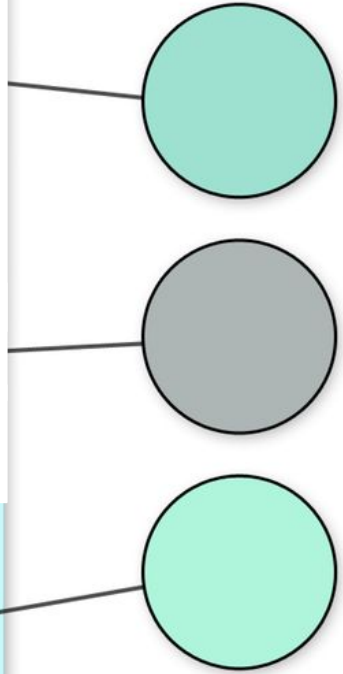


The case of the dress



Why is the brain doing this???



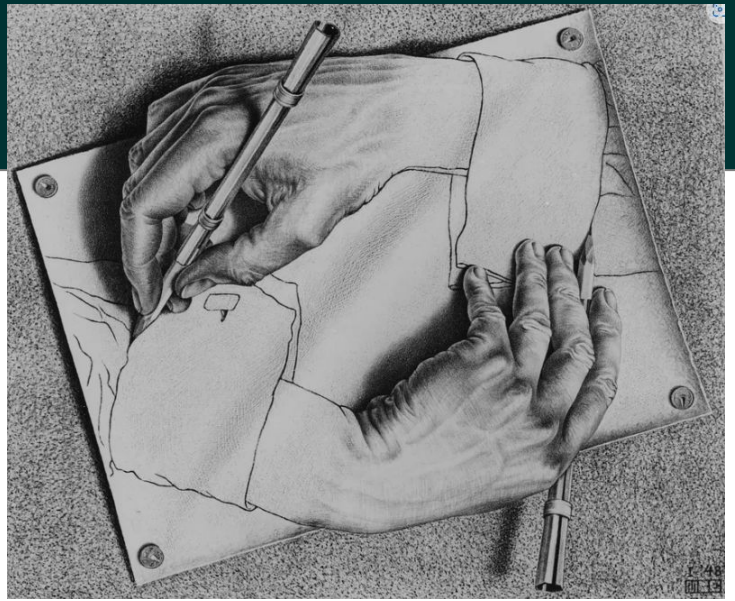


Why is the brain doing this?

Natural signals: highly redundant

↓

Brain actively anticipates upcoming sensory input rather than passively registering it; this anticipation is built on the current context of stimuli *(e.g, light of day, shadows, previous knowledge and experience)*



M. C. Escher,
Drawing hands (1948)

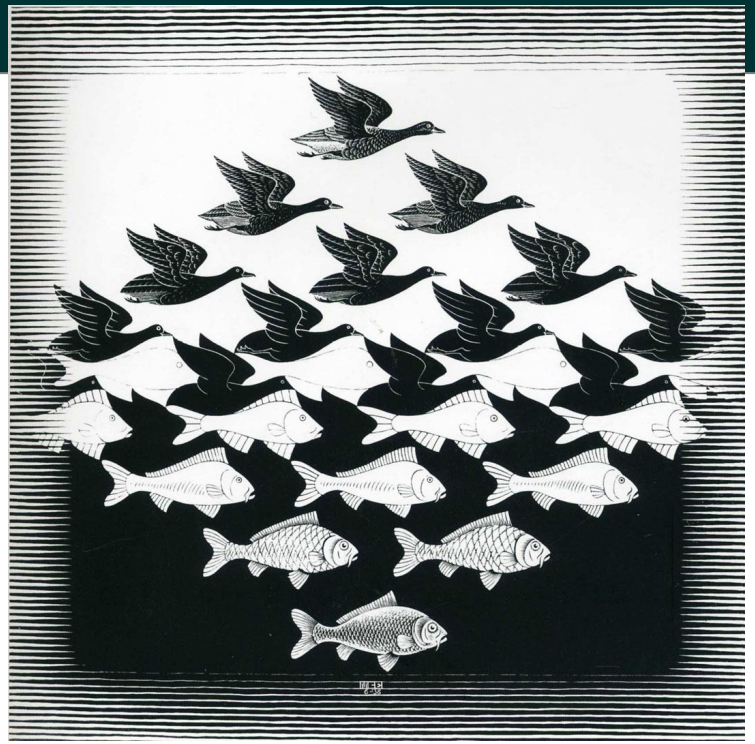
Goal: reduce redundancy and recode the sensory input into an efficient form

How is the brain doing this?

Predictive coding

Theoretical framework to explain how the brain processes information and generate perceptions of the world.

Hypothesis:
>> Brain operates as a predictive machine, where it is constantly making predictions of the world and comparing those with incoming sensory input in order to minimize mismatch errors.



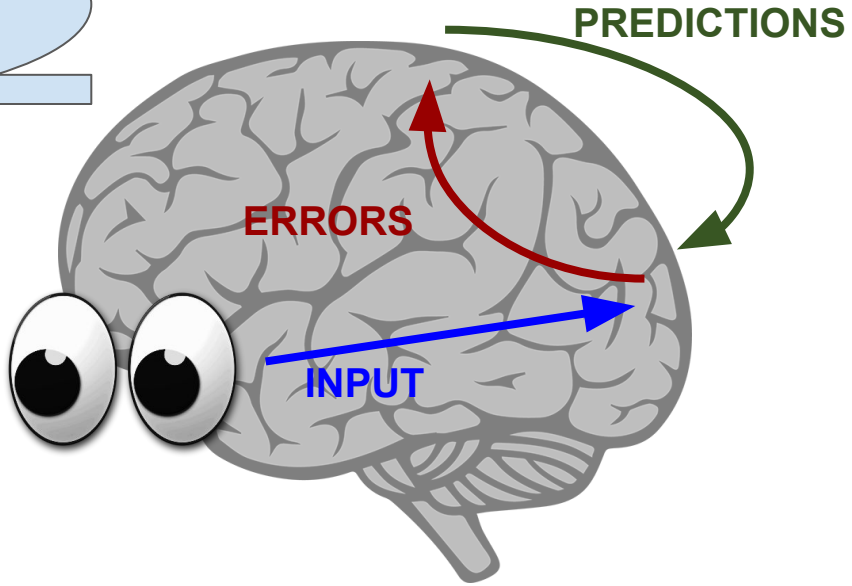
M. C. Escher,
Sky and water (1938)

How might predictive coding occur in the brain ?



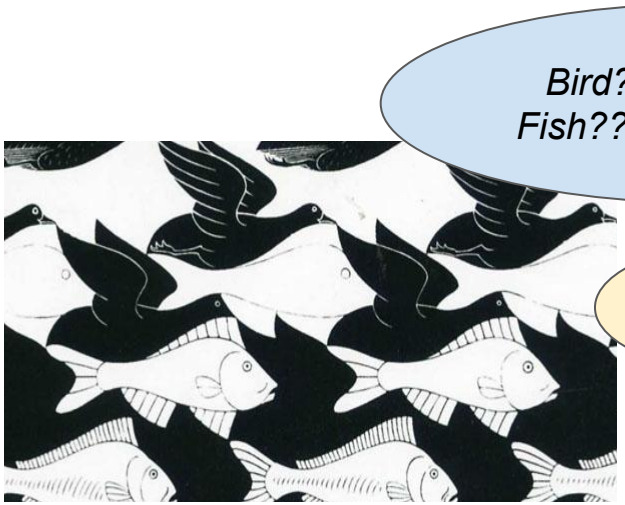
Bird? Fish? Bird???
Fish??? Birsh!?? Fird!?

Sugar???



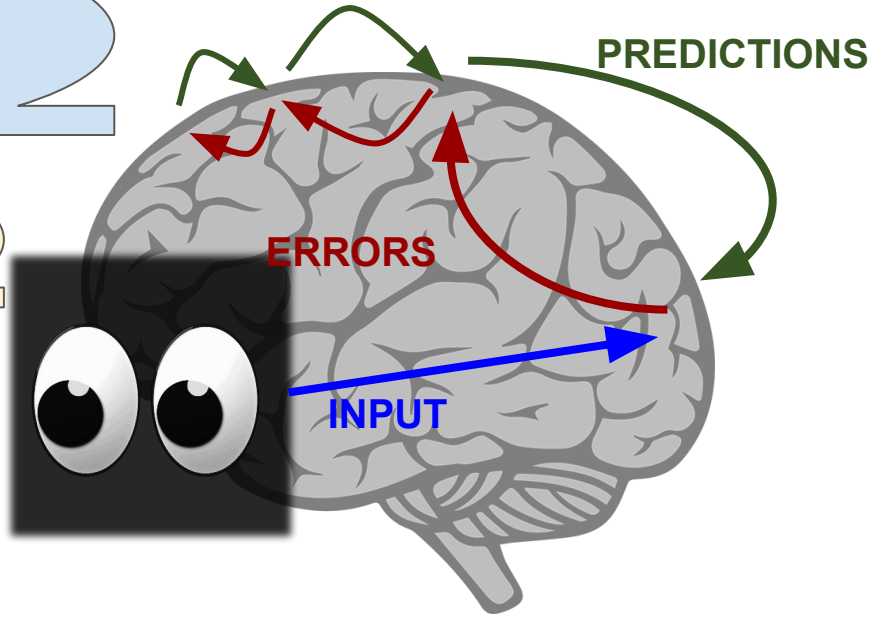
"A coffee with cream and DOG, please"

How might predictive coding occur in the brain ?



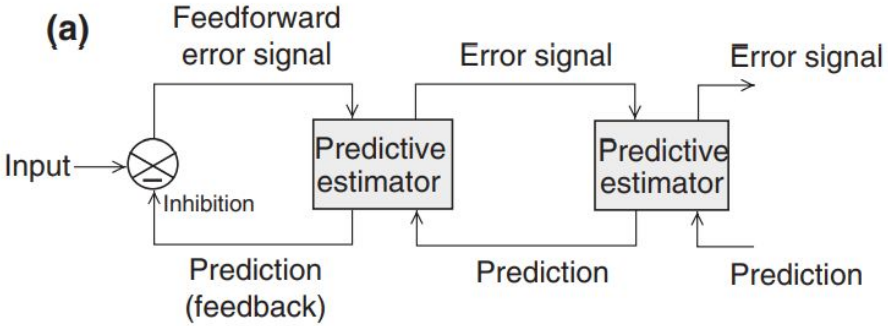
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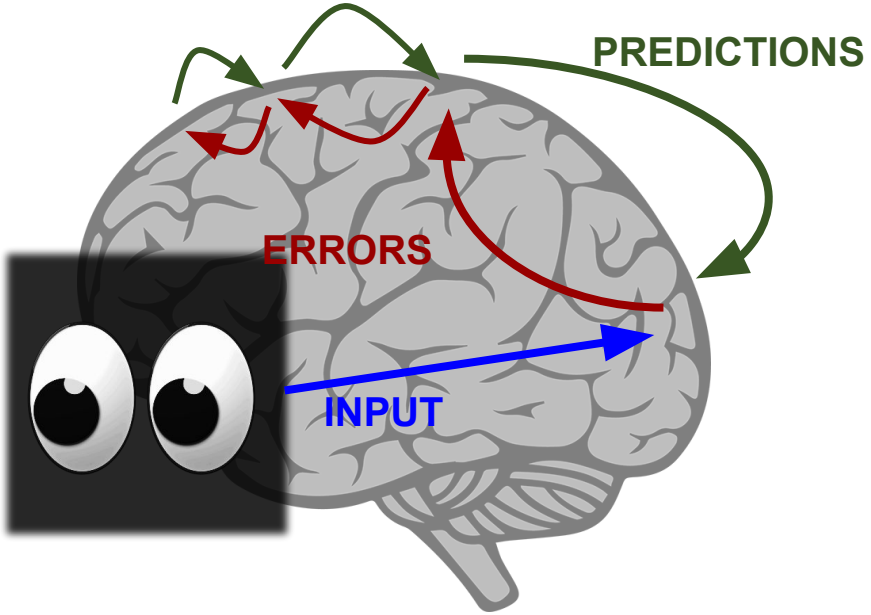
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How might predictive coding occur in the brain ?







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Goals of Fiser et al. 2016

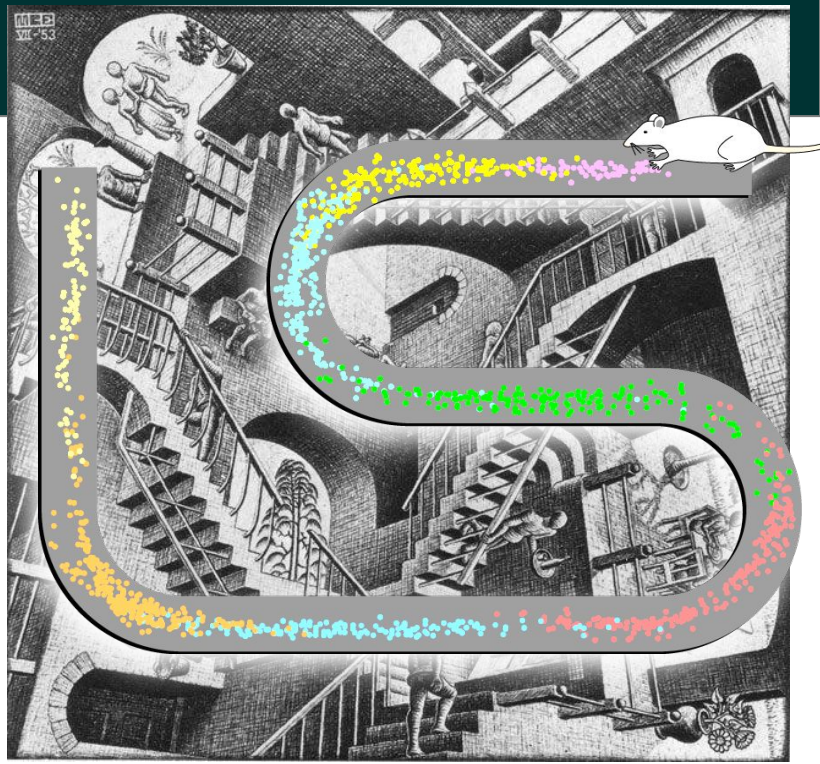
How do we navigate this world?

Internal representations of the environment

- Spatial maps in hippocampus and entorhinal cortex
- Strongly influenced by visual cues

How does this world navigate us?

Do these pre-established internal representations of the environment influence our sensory processing?



M. C. Escher,
Relativity (1953)

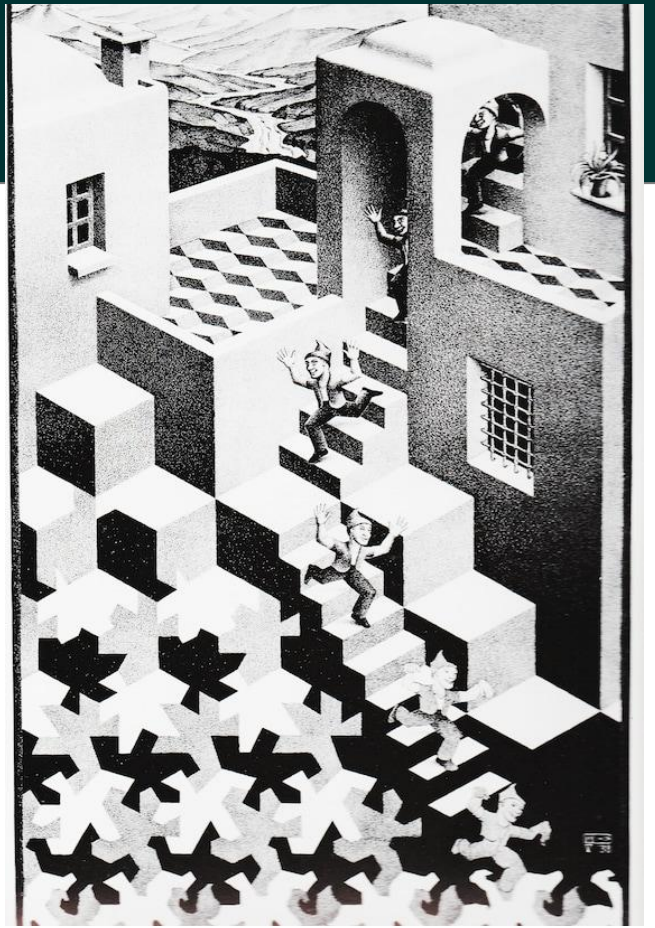
Goals of Fiser et al. 2016

Do these pre-established internal representations of the environment influence our sensory processing?

Hypothesis:
With experience, (internal) spatial maps influence visual processing, as spatial representations activate corresponding visual representations of the stimuli in specific locations of the environment.

- Which brain areas? Why?

  Areas involved in spatial memory that mediate top-down projections to visual cortex (ACC to V1)



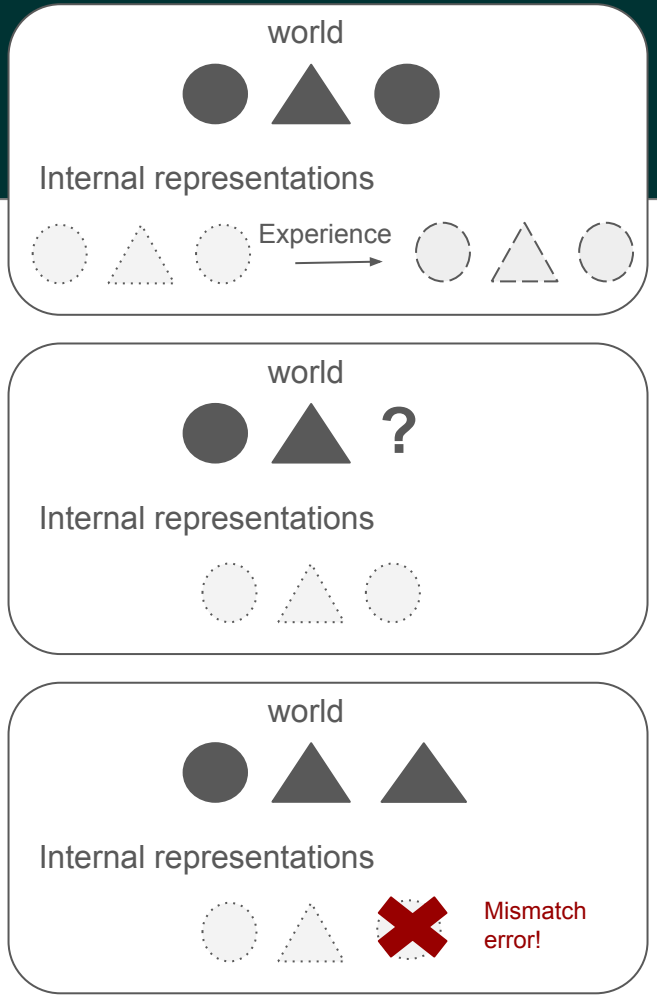
M. C. Escher, Cycle (1938)

Goals of Fiser et al. 2016

Do these pre-established internal representations of the environment influence our sensory processing?

Hypothesis:
With experience, (internal) spatial maps influence visual processing, as spatial representations activate corresponding visual representations of the stimuli in specific locations of the environment.

- What to be expected?
  1. Visual rep. should change with increasing experience;
  2. Predictive responses to an expected visual stimulus;
  3. Detectable mismatch (error) signals



Key methods

Experimental

- Behavioral task
- Two photon calcium imaging

Analysis

- Defining cell types based on selectivity
- Neural classifiers



M. C. Escher,
Reptiles (1943)

For each method:
background +
example

Behavioral task: background and example

Head-fixed subjects run through a virtual tunnel

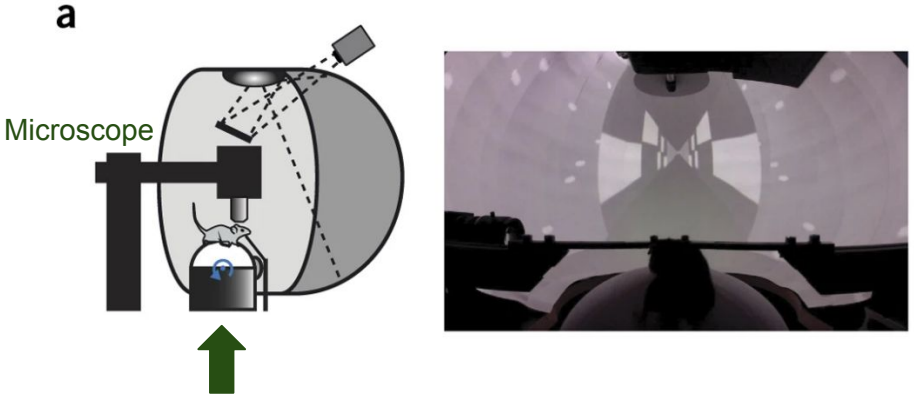


Fig. 1a

Behavioral task: background and example

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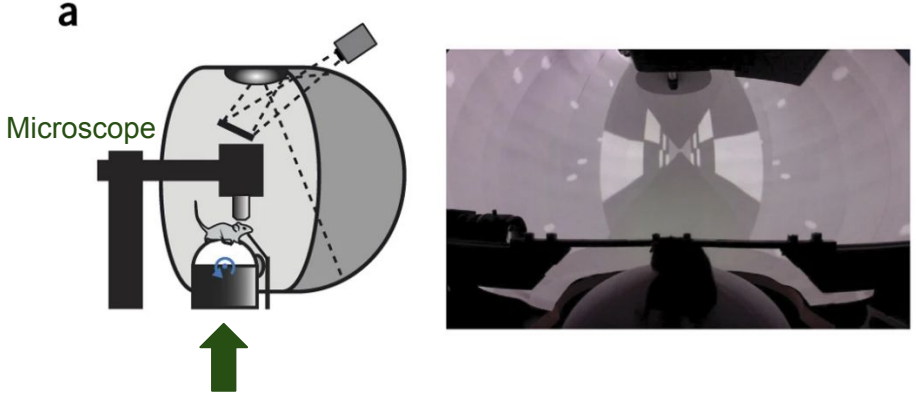
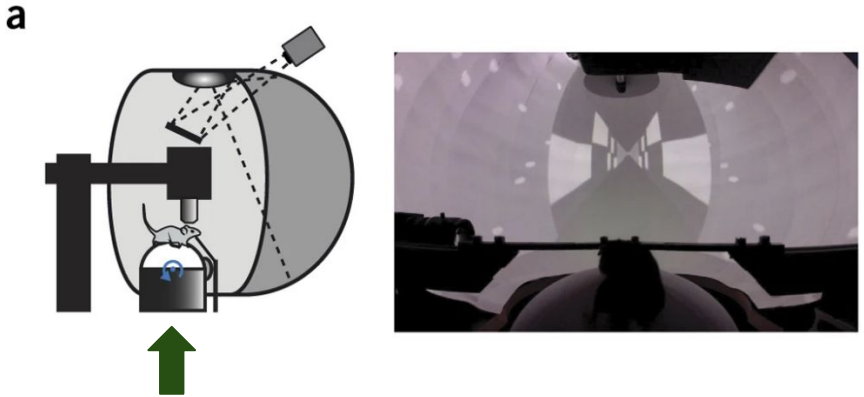


Fig. 1a

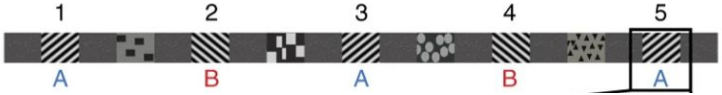
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Air-supported running ball
synced to visual display

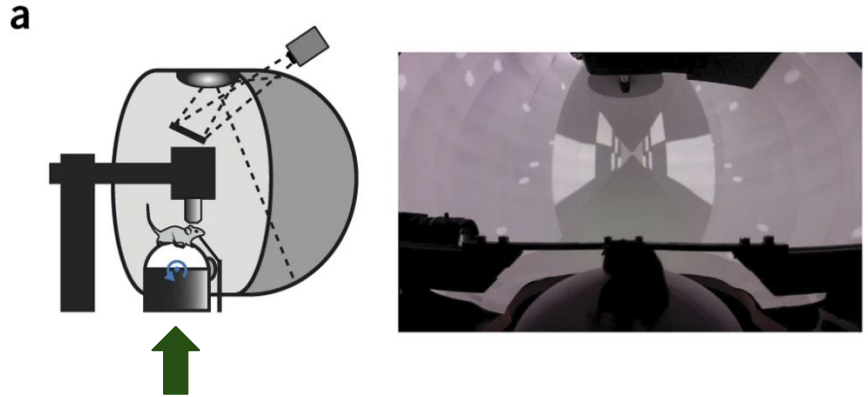
Tunnel design



- Two grating patterns: A and B

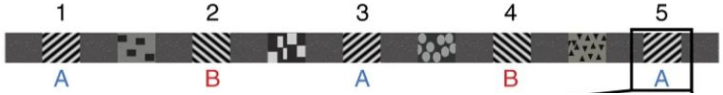
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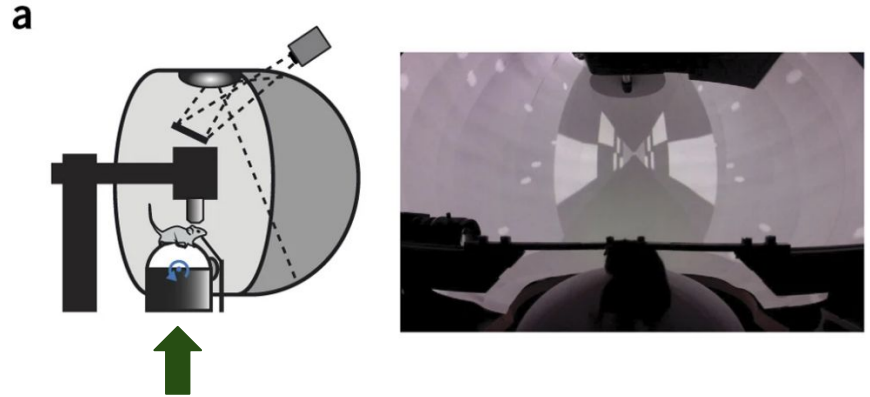
Tunnel design



- Two grating patterns: A and B
- A: positions 1 and 3, B: positions 2 and 4

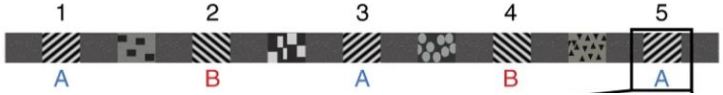
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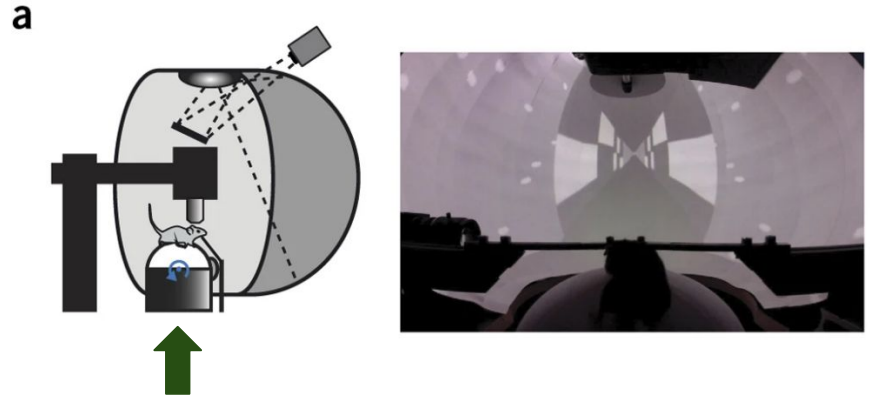
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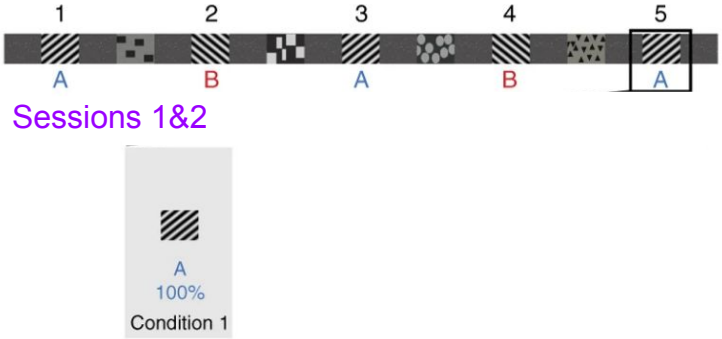
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Behavioral task: background and example

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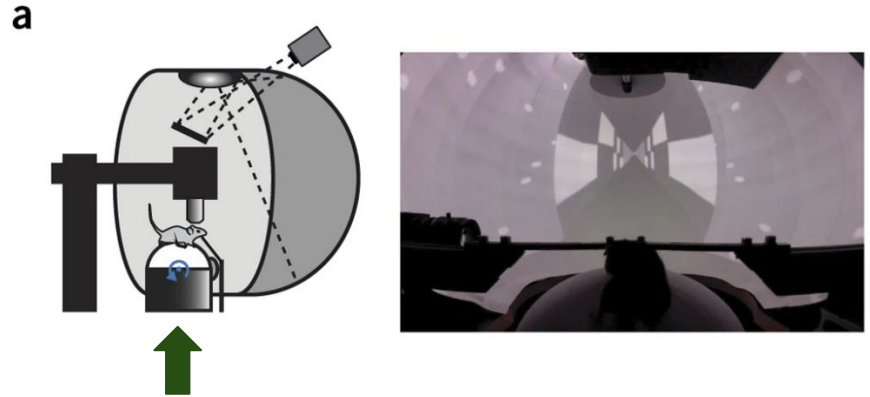
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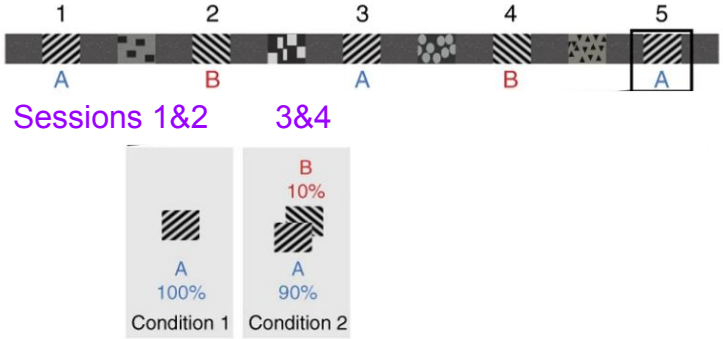
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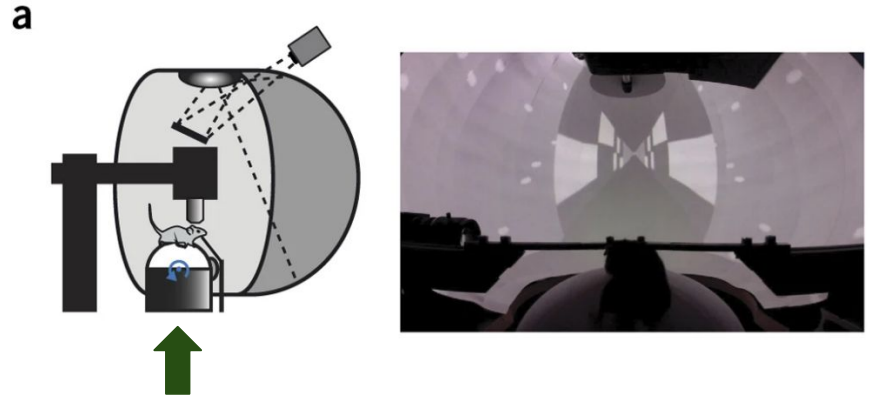
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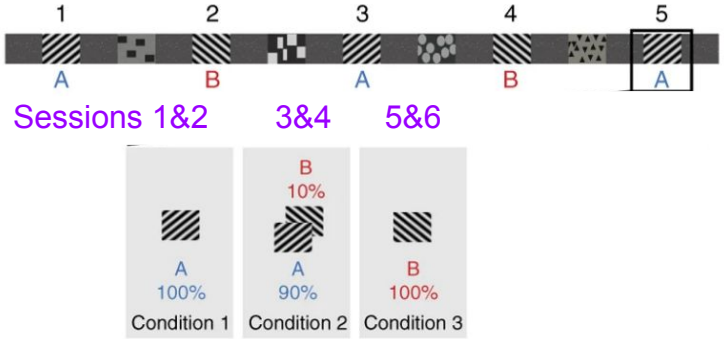
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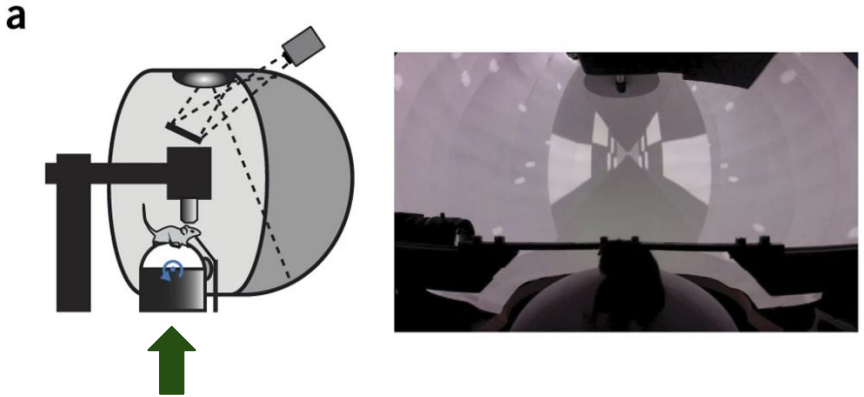
Tunnel design



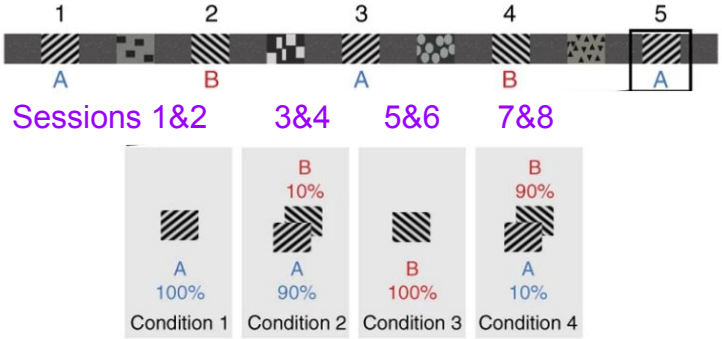
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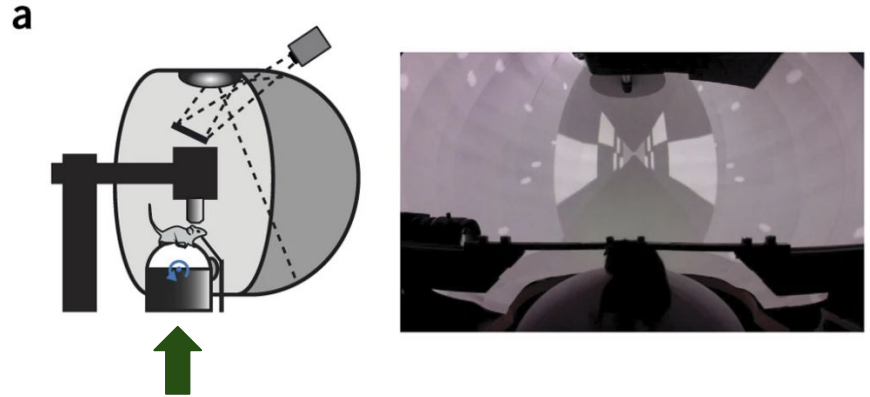
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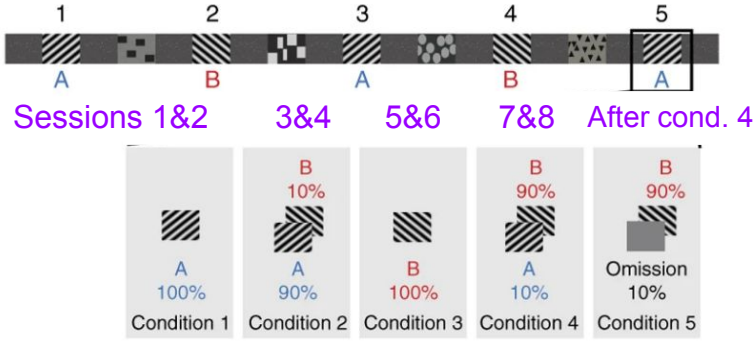
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Air-supported running ball synced to visual display

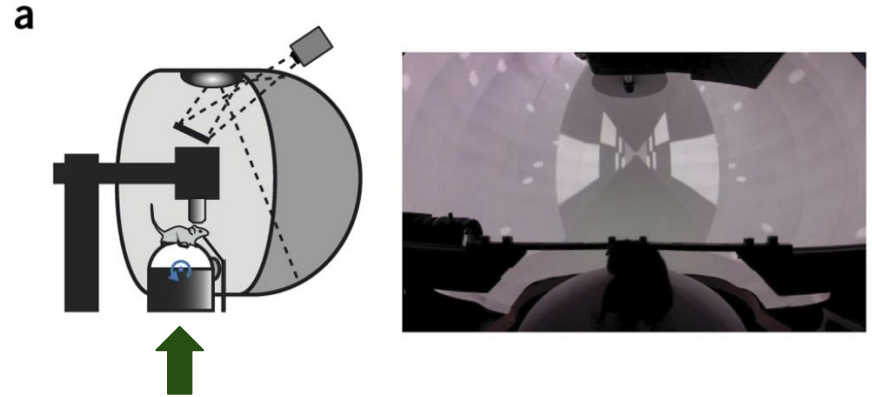
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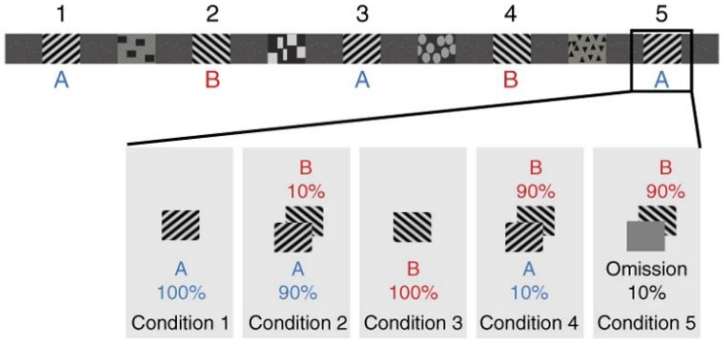
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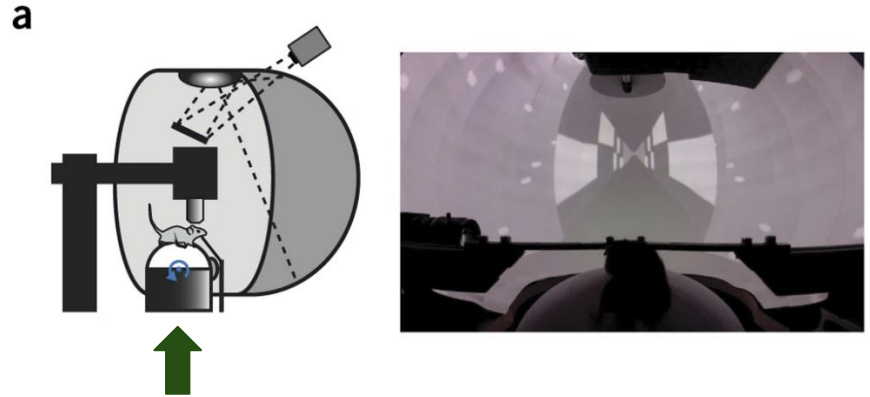
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- Subjects perform many runs (“trials”, av. 109/session)

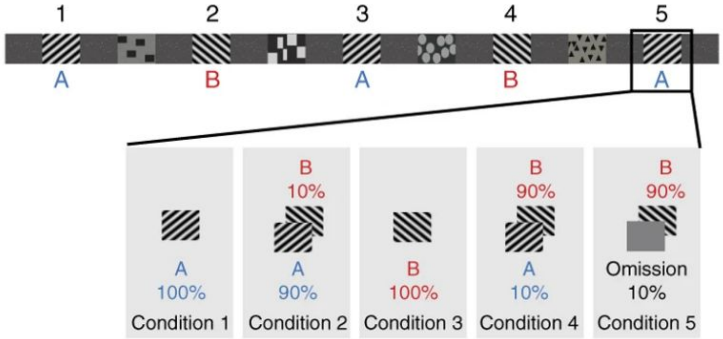
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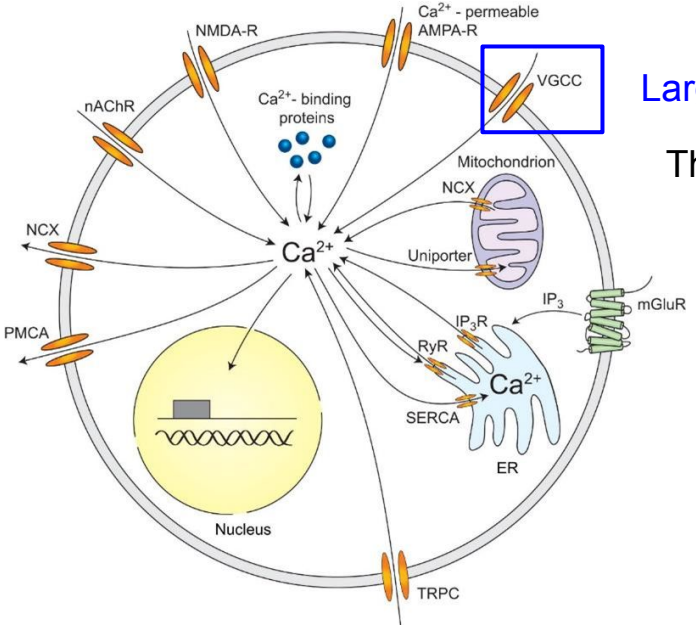
- Two grating patterns: A and B
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- Subjects perform many runs (“trials”, av. 109/session)
- **Why use**: How are subjects’ spatial experiences and predictions represented in primary visual cortex (V1)?

Two photon calcium imaging: background

When measuring neural activity, we have some options:

- Direct: ionic currents (electrophysiology)
- Indirect: Calcium (Ca^{2+} imaging). **Why Ca^{2+}?**

Ca^{2+} flow in neurons



Large Ca^{2+} inflow during spiking

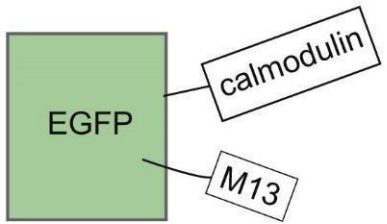
Therefore, can use Ca^{2+} as a proxy of neural activity

How do we measure Ca^{2+}?

Two photon calcium imaging: background

Genetically express a calcium indicator

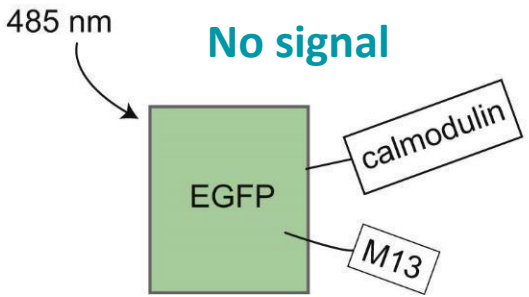
- This paper: GCaMP6f
- Converts Ca^{2+} to fluorescent signal (detect with microscope)



Two photon calcium imaging: background

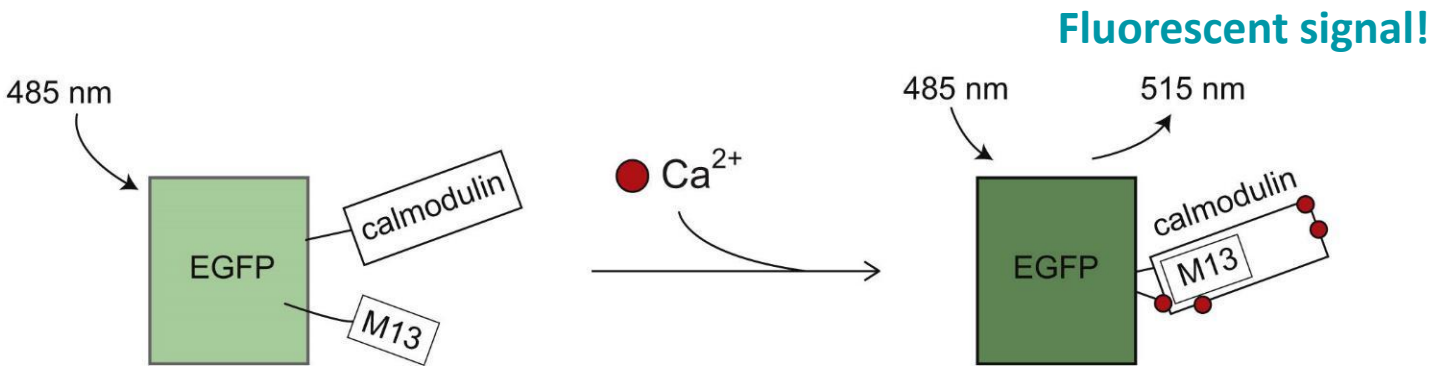
- This paper: GCaMP6f
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No Ca^{2+} present

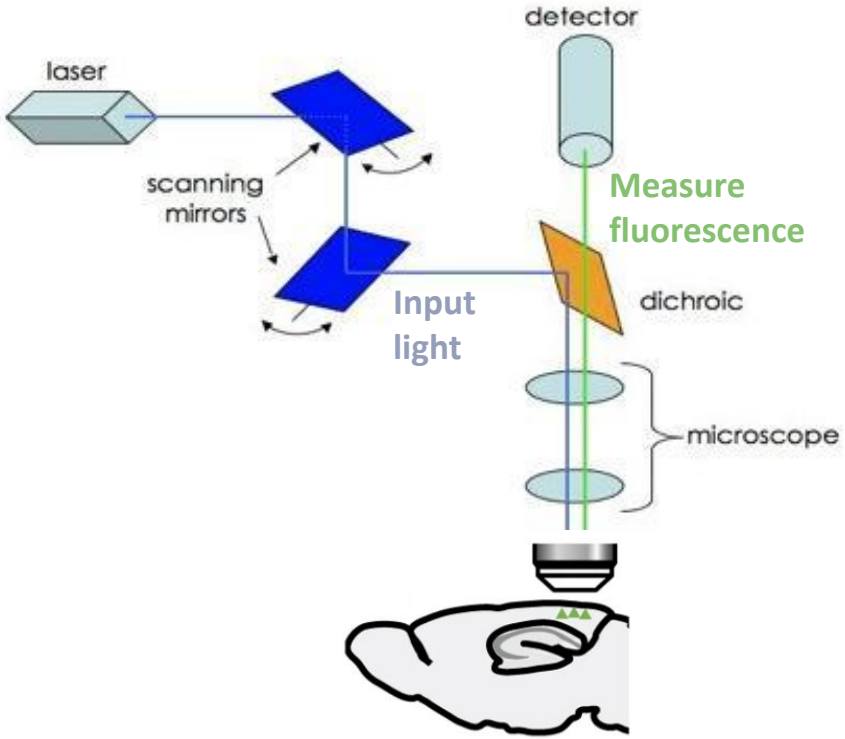


Two photon calcium imaging: background

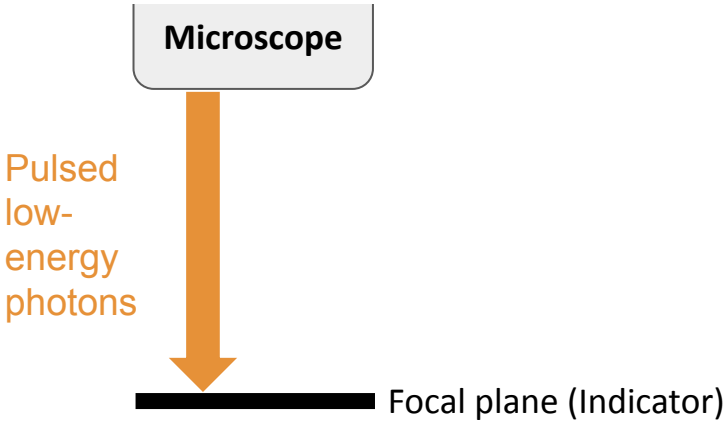
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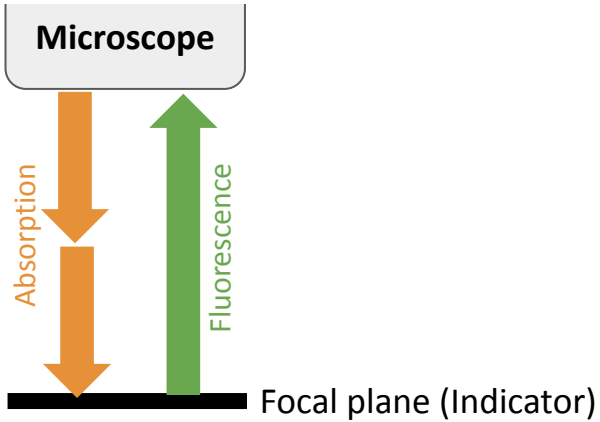


Two photon calcium imaging: background

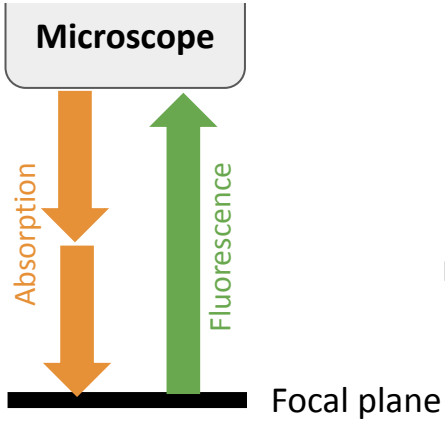


Two photon calcium imaging: background

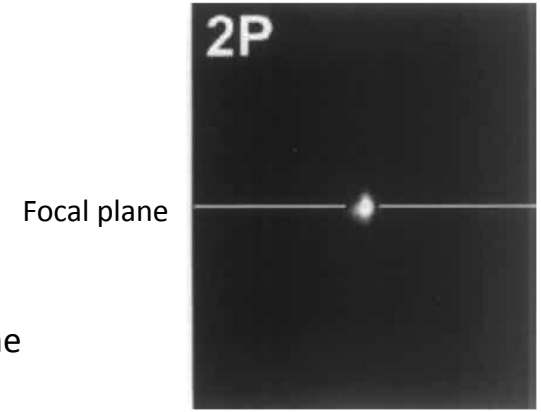
2 Low-energy photons absorbed simult.



Two photon calcium imaging: background



Example fluorescence measurements from 2 photon imaging



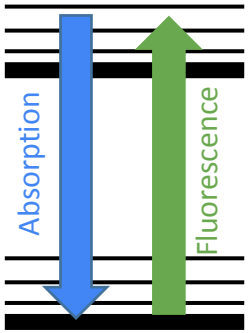
Fluorescence only from the focal plane

Why use: High resolution images at focal plane

Rubart *et al.* 1999

Two photon calcium imaging: background

One photon imaging

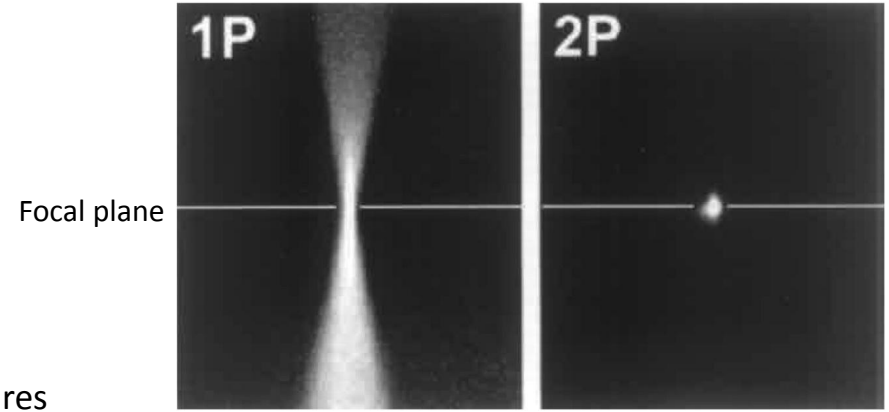


Two photon imaging



Indicator excitation requires almost simultaneous absorption of 2 photons (only occurs in focal plane)

Example fluorescence measurements from 1 vs 2 photon imaging



Fluorescence only from the focal plane

Why use: High resolution images at focal plane

Rubart *et al.* 1999

Two photon calcium imaging in V1: example

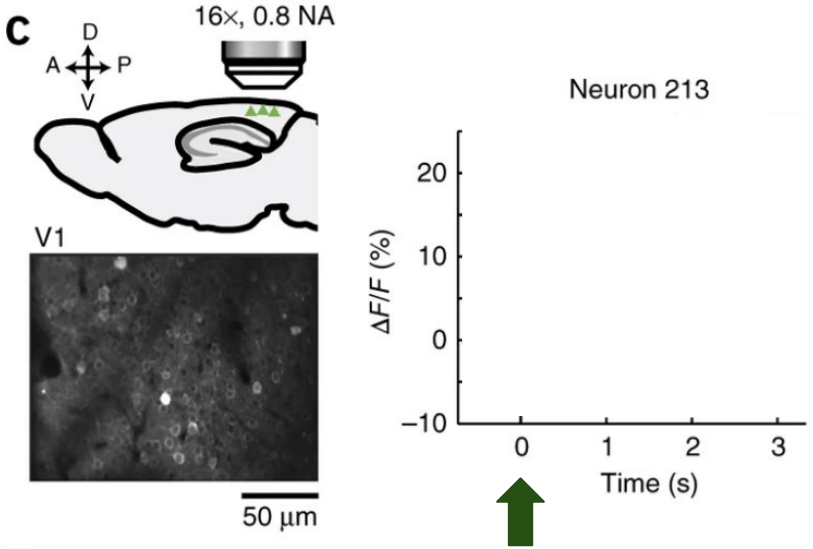


Fig. 1c-d

Two photon calcium imaging in V1: example

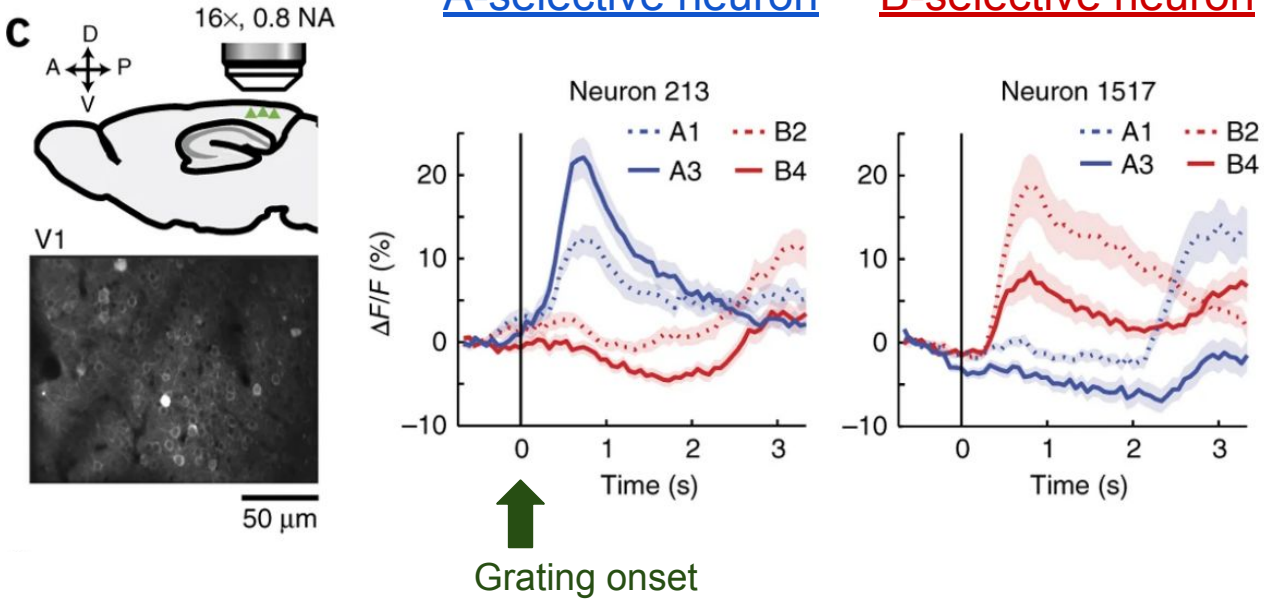
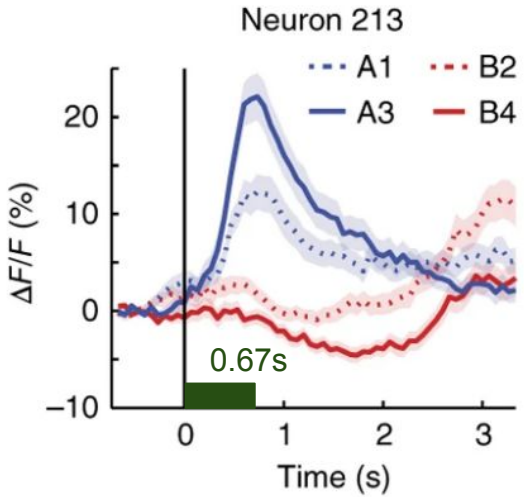


Fig. 1c-d
(mean +/- sem across trials)

Defining cell types: Selective cells

A-selective neuron



Each neuron

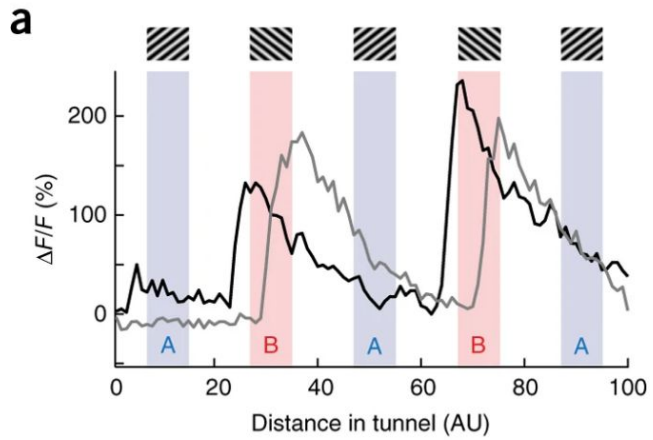
$$\text{Selectivity index (SI)} = \frac{(R_A - R_B)}{(R_A + R_B)}$$

- R_A: mean response to A in positions 1 & 3
- R_B: mean response to B in positions 2 & 4
- **Cell labeled as selective if abs(SI) > 0.5**

Defining cell types: Predictive and visual neurons

Defining cell types: Predictive and visual neurons

Activity traces of two neurons (**1** & 2) on a single trial (Fig. 2a)



- Activity of neuron 1 begins rising **before** B grating
- Activity of neuron 2 begins rising **during** B grating
- Both neurons show higher activity around B compared to A (selective)
- **Neuron 1: B-predictive neuron**
- **Neuron 2: B-visual neuron**
- Neurons can also be A-predictive, A-visual, or none of these

Defining cell types based on selectivity: Background & example

Predictive neurons (A or B)

- **Peak** within 0.333s of grating onset & **selective** (window: 0.133s before grating onset to 0.333s after grating onset)

Visual neurons (A or B)

- **Peak** after 0.533s of grating onset & **selective** (window: 0.533s to 1s after grating onset)

Example B-predictive and visual neurons

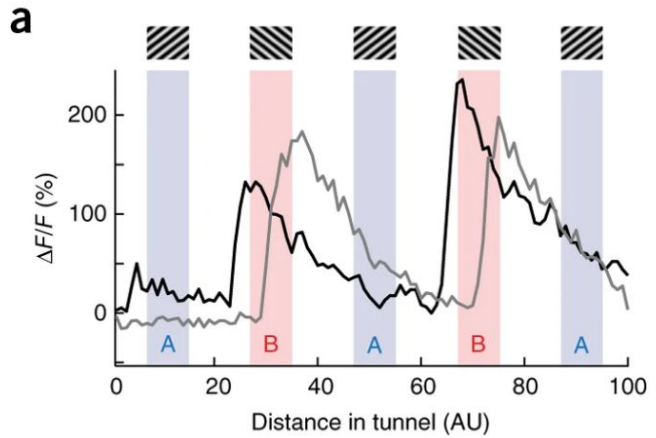
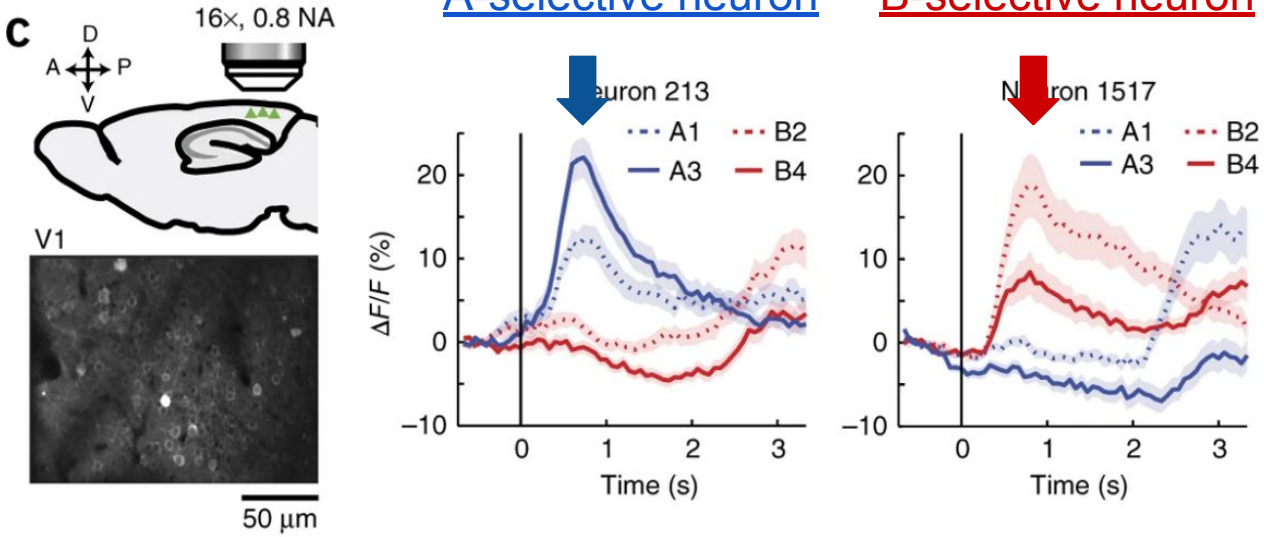


Fig 2a
Single trial

Two photon calcium imaging in V1: example



Neurons may contain information about grating type (A or B) **and** location

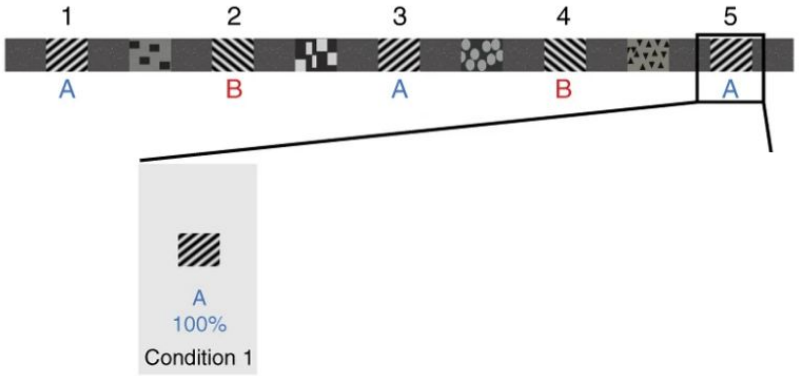
Fig. 1c-d
(mean +/- sem across trials)

Neural classifiers: Background

Does neural activity represent both grating type (A or B) and location?

Neural classifiers: Background

Does neural activity represent both grating type (A or B) and location?



3-dim. array of neural activity

A1, B2, A3, B4, A5

Neurons

Trials

Each trial: subject encounters A1, B2, A3, B4, A5

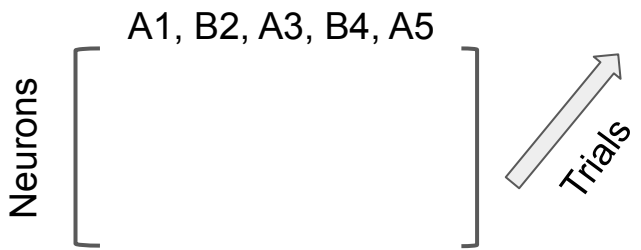
Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

Train & test classifiers (1 per condition)

- Split trials into training & testing datasets
- Using training dataset, train classifier (bagged decision trees) to predict grating (A1, B2, A3, B4, A5) based on neural data
- Test classifier on testing dataset

3-dim. array of neural activity



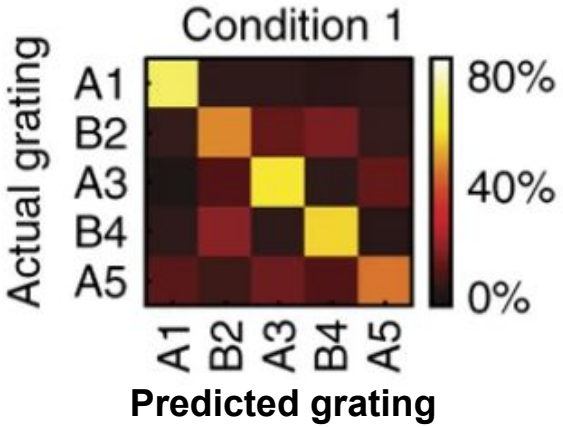
Neural classifiers: Background

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Presenting test results: Confusion matrix

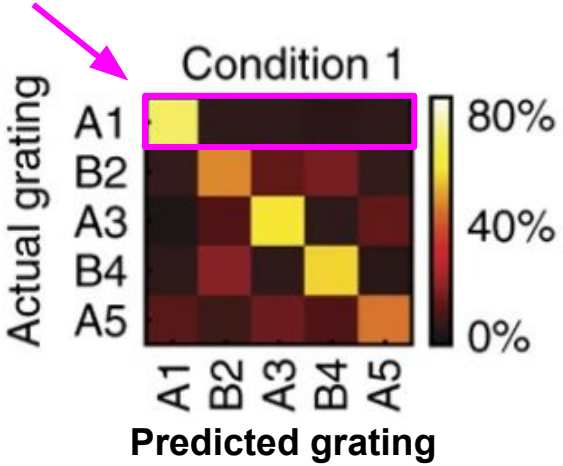


Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

- Of all actual A1 instances in testing dataset, what proportions are predicted as A1 (correct), B2, A3, B4, A5? (row sums to 1)
- Same idea for other rows
- **High performance <-> Strong diagonal: neural activity represents both grating type and location**
- Performance measure: mean over diagonal

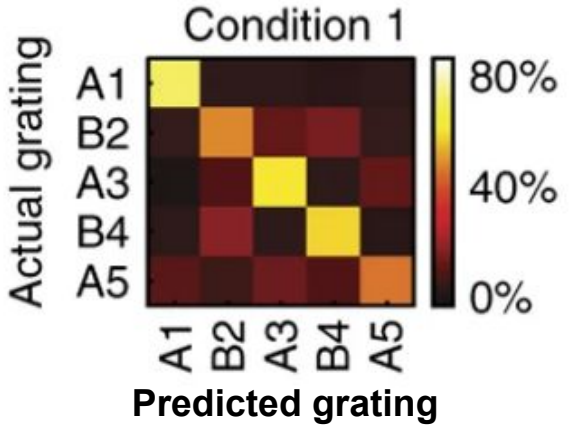
Presenting test results: Confusion matrix



Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

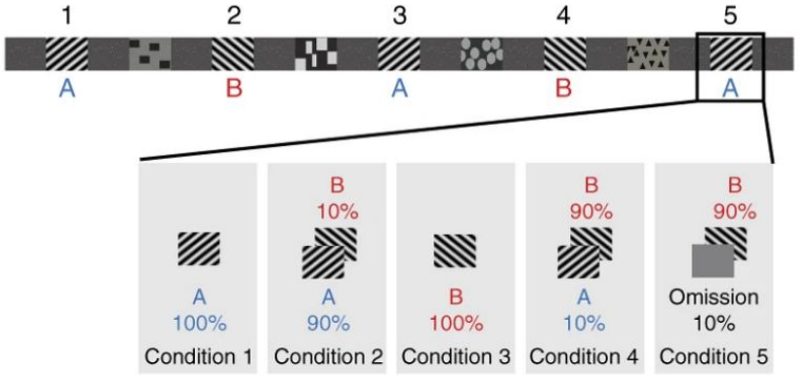
Train/test classifier for <u>each</u> condition

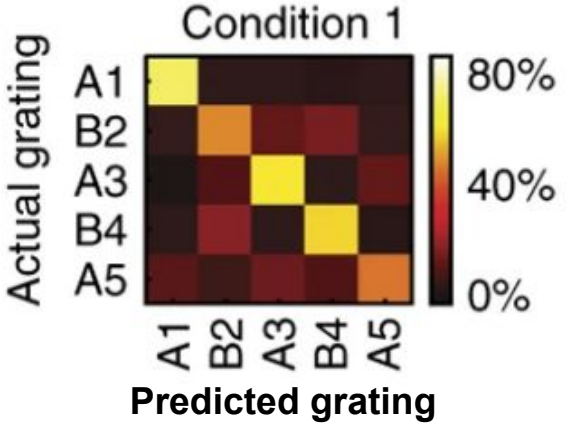


Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

Train/test classifier for <u>each</u> condition

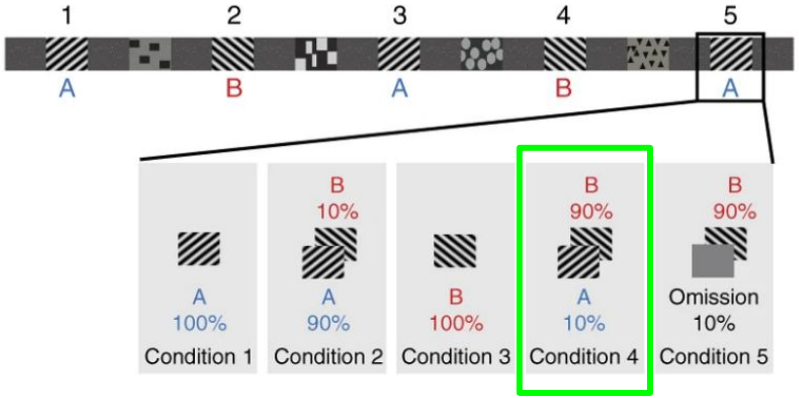




Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

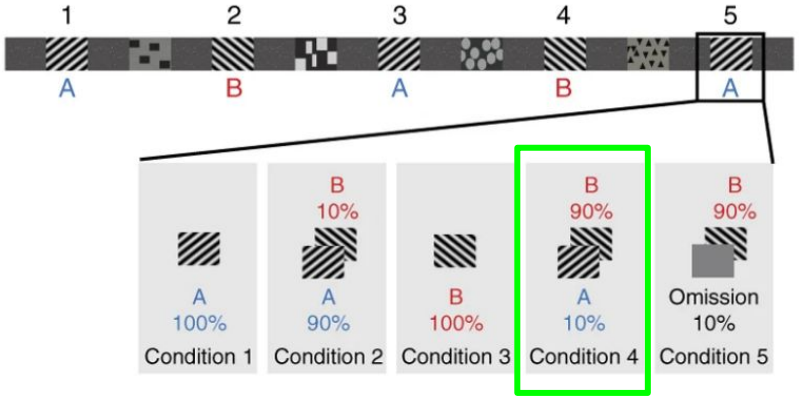
Train/test classifier for <u>each</u> condition

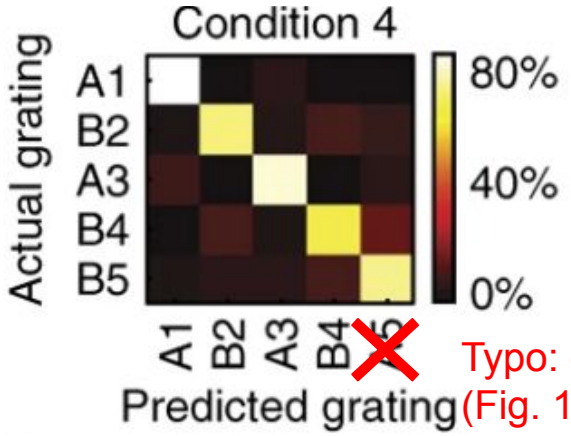


Neural classifiers: Background

Does V1 activity represent both grating type (A or B) and location?

Train/test classifier for <u>each</u> condition





Typo: correct B5 (Fig. 1e, bottom)

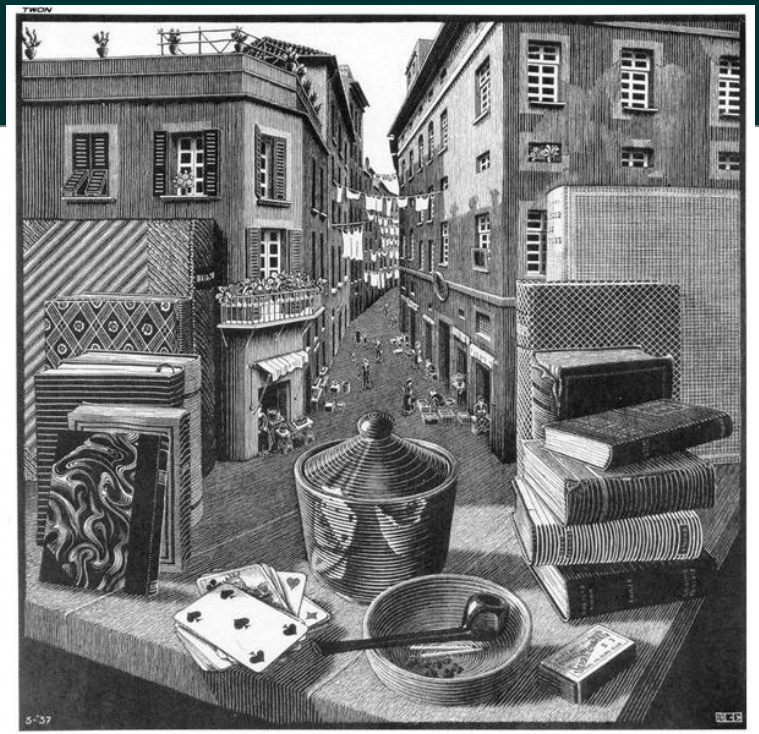
Homework instructions

- Read Fiser et al. 2016 (“Experience-dependent spatial expectations in mouse visual cortex”, see Moodle for paper)

- **By 11:59 pm on Tuesday 22nd Sept.:** Upload answers to “Exercise 1.1 Upload” on Moodle

Key notes:

- Late submissions will not be graded
- Answers with copied text from the paper or peers will not be graded
- *Tip: pay attention to grammar!*



M. C. Escher,
Still life and street (1937)